Omni University

Department of Computational Design

OmniLaTeX Thesis Template

Katherine Johnson

Submitted in partial fulfillment of the requirements for the degree of Master of Science in Engineering

Advisor: Prof. Alan Turing
Committee: Dr. Ada Lovelace, Dr. Donald Knuth
Defense Date: June 12, 2025

Hamburg, Germany
May 9, 2026

Rest of this page intentionally left blank.

Contents

Rest of this page intentionally left blank.

1. Introduction

Summarize the research motivation and objectives [Example 2024]. Provide context by reviewing relevant prior work and clearly state the gap this thesis aims to address. Outline the structure of the document and highlight the key contributions.

Rest of this page intentionally left blank.

2. Background and Related Work

Review the theoretical foundations underpinning the research [Knuth 1986]. Discuss established methods in the field, their strengths, and their limitations. Position the present work within the broader research landscape.

Rest of this page intentionally left blank.

3. Methodology

Document experimental design, data acquisition, and analysis steps [Knuth 1986]. The governing equation for the system under study is given by

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u + f(\mathbf{x}, t), \tag{3.1}$$

where u is the field variable, α is the diffusivity, and $f(\mathbf{x}, t)$ represents the source term. Numerical experiments were carried out on a uniform grid with spacing $\Delta x = 0.01$ and a time step satisfying the CFL condition $\Delta t \leq \Delta x^2/(2\alpha)$. Convergence was verified by refining the mesh until the L^2-norm of the error fell below 10^{-6}.

Rest of this page intentionally left blank.

4. Results

Present findings supported by figures and tables [Einstein 1905]. Figure 4.1 shows the convergence behaviour of the proposed scheme.

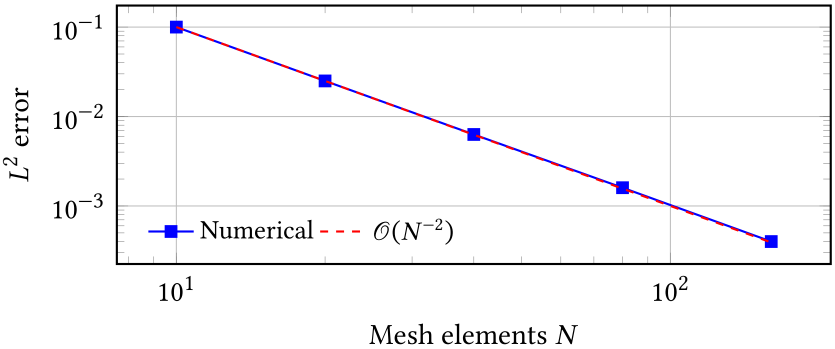

Figure 4.1: Convergence of the numerical solution with mesh refinement..

Table 4.1 summarises the error metrics for each refinement level.

Table 4.1: Error metrics for successive mesh refinements..

Level	Δx	L^2 error	Order
1	0.100	3.42×10^{-2}	—
2	0.050	8.67×10^{-3}	1.98
3	0.025	2.18×10^{-3}	1.99
4	0.012	5.47×10^{-4}	2.00

Rest of this page intentionally left blank.

5. Discussion

Interpret the results in the context of the research questions. Compare the observed convergence rate with theoretical predictions and discuss possible sources of discrepancy [Bird et al. 2009].

Rest of this page intentionally left blank.

6. Conclusion

Highlight contributions, limitations, and future work. Suggest directions for extending the methodology to more complex geometries and multi-physics coupling scenarios.

Rest of this page intentionally left blank.

A. Additional Material

Provide supporting derivations, extended data tables, and supplementary figures.

Rest of this page intentionally left blank.

Bibliography

BIRD et al. 2009 — BIRD, Steven et al. (2009): *Natural Language Processing with Python.* 1st ed. Beijing ; Cambridge [Mass.]: O'Reilly. 479 pp. ISBN: 978-0-596-51649-9 (cit. on p. 9).

EINSTEIN 1905 — EINSTEIN, Albert (1905): "Zur Elektrodynamik Bewegter Körper. (German) [On the Electrodynamics of Moving Bodies]." In: *Annalen der Physik* 322.10 (10), pp. 891–921. DOI: `http://dx.doi.org/10.1002/andp.19053221004` (cit. on p. 7).

EXAMPLE 2024 — EXAMPLE, Jane (2024): *Demonstration Citation Entry.* Placeholder reference used for OmniLaTeX citation examples (cit. on p. 1).

KNUTH 1986 — KNUTH, Donald Ervin (1986): *The TeXbook.* Computers & Typesetting A. Reading, Mass: Addison-Wesley. 483 pp. ISBN: 978-0-201-13447-6 (cit. on pp. 3, 5).